

Solution Requirements Specification

Date:	3 October 2026	Team ID:	UNM1611SHARVESHAM
Project Name:	AI BlogNest API	Maximum Marks:	4 Marks

Functional and Non-Functional Requirements:

This document defines the comprehensive operational specifications for AI BlogNest API. Functional Requirements (FR) define what the system does, while Non-Functional Requirements (NFR) specify quality, performance, and security criteria.

Step 1: Functional Requirements (FR) — Part 1

S.No	FR Category	Requirement Description	Priority
1	Authentication	Users must securely register with name, unique email, and minimum 6-character password. Passwords are encrypted with bcrypt salt factor 10. Login issues a signed JWT token valid for 7 days.	High
2	Authorization Levels	Public users can read published blogs (GET /api/blogs, GET /api/blogs/:id). Authenticated users can create blogs and invoke AI generation. Only the original author can edit or delete their own posts.	High
3	External Interfaces	Integration with Google Gemini Generative AI API via @google/generative-ai SDK. Communicates with MongoDB Atlas cloud database via Mongoose ODM over TLS connection string.	High
4	Transactions Processing	Handles atomic CRUD database transactions: user document creation, blog document creation, idempotent query retrieval, atomic update, and deletion with cascading checks.	High

Functional Requirements (Part 2) & Non-Functional Requirements

Step 1: Functional Requirements (FR) — Part 2

S.No	FR Category	Requirement Description	Priority
5	Reporting & Querying	Endpoint GET /api/blogs returns list of blogs sorted descending by creation date, dynamically populating author name and email fields from the User collection.	Medium
6	Business Rules & AI Logic	- Generates structured 100-150 word blogs on given topic. - Regex engine sanitizes raw markdown/asterisks. - Auto-infers category across 9 domains (Tech, Health, etc.) if not provided.	High
7	Compliance & Privacy	Credentials never stored in plaintext; API secrets stored in .env and ignored by git; strict request payload validation via express-validator to prevent injection attacks.	High
8	AI Summarization	Endpoint POST /api/ai/summarize accepts arbitrary blog content and generates a single crisp executive summary paragraph using Gemini AI.	Medium

Step 2: Non-Functional Requirements (NFR) — Part 1

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast response times for API queries and external LLM dispatch.	CRUD latency < 100 ms; Gemini generation < 2.5 seconds
2	Scalability	Stateless architecture capable of handling concurrent user requests.	Supports 500+ concurrent connections; horizontal scaling ready
3	Security & Privacy	Cryptographic password hashing and tamper-evident JWT authorization.	Bcrypt salt=10; HMAC-SHA256 tokens; OWASP-compliant sanitized inputs

Non-Functional Requirements (Part 2) & Traceability Matrix

Step 2: Non-Functional Requirements (NFR) — Part 2

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
4	Reliability & Uptime	Robust server uptime with automatic MongoDB reconnection and centralized error recovery.	99.9% availability; zero uncaught exceptions crashing the Node process
5	Usability & API Standards	Strict RESTful conventions with semantic HTTP status codes (200, 201, 400, 401, 403, 404, 422, 500) and JSON responses.	100% compliant with REST standards; Thunder Client / Postman collections provided
6	Maintainability	Clean MVC architecture separating routes, controllers, models, and services for straightforward maintenance.	Modular codebase; zero circular dependencies; Prettier formatting

Requirements Traceability & Verification Matrix:

Req ID	Feature Description	Source File / Implementation	Verification Method	Status
FR-01	User Registration & Bcrypt	authController.js / User.js	Thunder Client: POST /register	Verified
FR-02	JWT Login & Bearer Token	authController.js / authMiddleware.js	Thunder Client: POST /login	Verified
FR-03	Blog CRUD Operations	blogController.js / Blog.js	Thunder Client: /api/blogs suite	Verified
FR-04	Gemini Blog Post Generation	aiController.js / geminiService.js	Thunder Client: POST /generate-blog	Verified
FR-05	Gemini Blog Summarization	aiController.js / geminiService.js	Thunder Client: POST /summarize	Verified
FR-06	Category Auto-Classification	aiController.js (inferCategory)	Unit test across 9 keyword sets	Verified